

PATENT LANDSCAPE REPORT: HYDROGENATED NBR SYNTHESIS, DEGRADATION, AND POST-TREATMENT TECHNOLOGIES

1. ABSTRACT

This report synthesizes nine patents relevant to hydrogenated nitrile butadiene rubber (HNBR), with emphasis on routes that create, modify, or purify hydrogenated NBR materials. The target material is defined as nitrile rubber derived from conjugated dienes and α,β -unsaturated nitriles that has been partially to highly hydrogenated to reduce residual unsaturation and improve heat, ozone, oil, and aging resistance. Across the patent set, two major technology directions emerge. First, several patents focus on controlled molecular-weight reduction of NBR prior to hydrogenation by olefin metathesis using ruthenium catalysts, co-olefins such as ethylene or 1-hexene, and solution-phase processing in monochlorobenzene. These routes aim to produce lower-Mooney, narrower-distribution HNBR suitable for seals, hoses, and other molded articles. Second, other patents address formulation and post-processing of HNBR, including partially hydrogenated NBR blended with silicone rubber, peroxide-cured HNBR compositions containing microgels and unsaturated carboxylic acids, and catalyst-residue removal using functionalized ion-exchange resins. A separate catalyst-system patent shows that alkaline-earth metal chlorides can accelerate metathesis of NBR, while another demonstrates that salt additives can strongly influence metathesis efficiency and molecular-weight reduction. The landscape also includes a metathesis-catalyst patent that explicitly claims modified NBR depolymerization and hydrogenation uses, underscoring the centrality of ruthenium carbene catalysts for HNBR precursor modification. Overall, the evidence points to solution metathesis as the dominant precursor-engineering strategy, with hydrogenation typically performed using Wilkinson-type Rh catalysts or related homogeneous systems. Process trends include high-temperature solution reactions, inert-atmosphere handling, catalyst loading optimization, and downstream purification to reduce residual Ru. Formulation trends emphasize balancing low-temperature flexibility, compression set, and high-temperature durability through selective hydrogenation level control and reinforcing or compatibilizing additives.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7919563B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a two-step route in which nitrile rubber is first subjected to olefin metathesis in solution to reduce molecular weight, then hydrogenated to form low-molecular-weight HNBR. The metathesis uses ruthenium catalysts and co-olefins such as ethylene or 1-hexene in monochlorobenzene, followed by hydrogenation with Wilkinson's catalyst under high hydrogen pressure. - Cement concentration: 6 or 15% by weight — Examples 1-4 - Agitator speed: 600 rpm — Examples 1-4 - Solvent: monochlorobenzene — Examples 1-4 - Substrate acrylonitrile content: 34 mol % — Examples 1-4 - Substrate Mooney viscosity: 35 ML (1+4) @ 100 deg. C. — Examples 1-4 - Ethylene pressure: 6.9×10^5 Pa — Examples 1-3 - Nitrogen pressure: 100 psi — Example 4 - Co-olefin concentration: 500 psi — Table 1 - Reactor temperature: 80° C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Rubber mass: 200 g — Example 1 - MCB mass: 1133 g — Example 1 - Rubber mass: 75 g — Example 3 - MCB mass: 1175 g — Example 3 - 1-hexene mass: 150 g — Example 4 - Hydrogen pressure: 1200 psi — hydrogenation conditions - Hydrogenation temperature: 138° C. — hydrogenation conditions - Wilkinson's catalyst loading: 0.08 phr — hydrogenation conditions - Triphenylphosphine: 1 phr — hydrogenation conditions - Hydrogen degassing: 3 times with H₂ (100 psi) — hydrogenation procedure - Hydrogenation preheat: 130° C. — hydrogenation procedure - Reaction monitoring: residual double bond (RDB) by IR spectroscopy — hydrogenation procedure - Mn: 27 kg/mol — typical product - Mw: 54 kg/mol — typical product - Starting polymer Mn: 85 kg/mol — typical product - Starting polymer Mw: 296 kg/mol — typical product - MWD/PDI: 3.4 to 2.0 — typical product comparison - PDI minimum: 1.90 — Table 2 - Mooney viscosity: 73 to 3 — Table 2

Relevant Experimental Evidence - Example 1: 15 wt.-% cement was degassed with ethylene at 6.9×10^5 Pa before metathesis. - Example 2: 15 wt.-% cement was processed under the same ethylene degassing conditions as Example 1. - Example 3: 6 wt.-% cement was used, showing the lower-solids metathesis variant. - Example 4: 1-hexene was used as co-olefin instead of ethylene, demonstrating alternative chain-scission conditions. - Examples 1-4: metathesis followed by hydrogenation gave lower Mn/Mw and narrower PDI than the starting polymer.

Technical Relevance This patent is directly relevant because it provides a controlled route to low-molecular-weight HNBR, a key target for improved processability. It also discloses practical hydrogenation conditions and shows how metathesis precursor engineering can tune final HNBR molecular weight and viscosity.

Patent 2

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent also teaches a two-step process in which nitrile rubber is first degraded by olefin metathesis in the presence of a ruthenium catalyst and a co-olefin, then hydrogenated to HNBR. The emphasis is on producing lower-molecular-weight, narrower-distribution HNBR and shaped articles from it. - Cement concentration: 6 or 15% by weight — Examples 1-4 - Agitator speed: 600 rpm — Examples 1-4 - Solvent: monochlorobenzene — Examples 1-4 - Substrate acrylonitrile content: 34 mol % — Examples 1-4 - Substrate Mooney viscosity: 35 ML (1+4) @ 100 deg. C. — Examples 1-4 - Ethylene pressure: 6.9×10^5 Pa — Examples 1-3 - Nitrogen pressure: 100 psi — Example 4 - Co-olefin concentration: 500 psi — Table 1 - Reactor temperature: 80° C. — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Rubber mass: 200 g — Example 1 - MCB mass: 1133 g — Example 1 - Rubber mass: 75 g — Example 3 - MCB mass: 1175 g — Example 3 - 1-hexene mass: 150 g — Example 4 - Hydrogen pressure: 1200 psi — hydrogenation conditions - Hydrogenation temperature: 138° C. — hydrogenation conditions - Wilkinson's catalyst loading: 0.08 phr — hydrogenation conditions - Triphenylphosphine: 1 phr — hydrogenation conditions - Hydrogen degassing: 3 times with H₂ (100 psi) — hydrogenation procedure - Hydrogenation preheat: 130° C. — hydrogenation procedure - Reaction monitoring: residual double bond (RDB) by IR spectroscopy — hydrogenation procedure - Mn: 27 kg/mol — typical product - Mw: 54 kg/mol — typical product - Starting polymer Mn: 85 kg/mol — typical product - Starting polymer Mw: 296 kg/mol — typical product - MWD/PDI: 3.4 to 2.0 — typical product comparison - PDI minimum: 1.90 — Table 2 - Mooney viscosity: 73 to 3 — Table 2

Relevant Experimental Evidence - Example 1: 15 wt.-% cement was degassed with ethylene at 6.9×10^5 Pa before metathesis. - Example 2: 15 wt.-% cement was processed under the same ethylene degassing conditions as Example 1. - Example 3: 6 wt.-% cement was used, showing the lower-solids metathesis variant. - Example 4: 1-hexene was used as co-olefin instead of ethylene, demonstrating alternative chain-scission conditions. - Examples 1-4: metathesis followed by hydrogenation gave lower Mn/Mw and narrower PDI than the starting polymer.

Technical Relevance This patent is highly relevant because it discloses the same low-molecular-weight HNBR concept with explicit process and property data. It supports the use of metathesis plus hydrogenation to tailor HNBR viscosity and molecular-weight distribution for improved processing.

Patent 3

- Patent Number: EP2580260A1
- Patent Title: Blends of partially hydrogenated nitrile rubber and silicone rubber, ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent is not a synthesis of HNBR itself but a blend technology using partially hydrogenated nitrile rubber and vinyl-containing silicone rubber. The blend is prepared in the presence of a metathesis catalyst, with an emphasis on solvent-free or very low-solvent processing and improved vulcanizate properties. - Organic solvent content: at most 1200 ppm — claim 2 - Organic solvent content: at most 500 ppm — claim 2 - Organic solvent content: at most 100 ppm — claim 2 - Compression set test compression: 25% — Examples 1.3/1.7 discussion - Compression set storage temperature: -30°C — Examples 1.3/1.7 discussion - Compression set storage temperature: 150°C — Examples 1.3/1.7 discussion - Compression set storage time: 70 h — Examples 1.3/1.7 discussion - Compression set storage time: 158 h — Examples 1.3/1.7 discussion - Compression set storage time: 168 h — Examples 1.3/1.7 discussion

Relevant Experimental Evidence - Examples 1.3, 1.5 and 1.7: inventive vulcanizates showed equal or higher stress values at 50% and 100% elongation than comparative examples. - Examples 1.3 and 1.7: lower compression set after -30°C/70 h and 150°C/168 h storage indicated improved low- and high-temperature performance. - The patent demonstrates metathesis-catalyzed blending of partially hydrogenated NBR with silicone rubber under low-solvent conditions.

Technical Relevance This patent is relevant because it uses partially hydrogenated NBR as a key component in high-performance elastomer blends. It highlights how hydrogenation level and formulation strategy affect compression set and thermal durability, which are important end-use properties of HNBR.

Patent 4

- Patent Number: US9321910B2
- Patent Title: Microgel-containing vulcanizable composition based on hydrogenated nitrile ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown

- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a vulcanizable HNBR composition containing hydrogenated nitrile rubber, peroxide, unsaturated carboxylic acid or salt, and microgels. The HNBR itself is prepared by hydrogenation of nitrile rubber, and the patent focuses on compounding and vulcanization behavior rather than the hydrogenation step. - Microgel polymerization temperature: 30° C. — microgel preparation - Microgel reaction temperature: 40° C. — microgel preparation - Microgel reaction temperature: 50° C. — microgel preparation - Polymerization conversion: >95% — microgel preparation - Diethylhydroxylamine: 23.5 g — microgel termination - Microgel A butadiene: 12608 g — Table 1 - Microgel A styrene: 287 g — Table 1 - Microgel A TMPTMA: 530 g — Table 1 - Microgel A HEMA: 137 g — Table 1 - Microgel A water: 250 g — Table 1 - Microgel A initiation: 3805.5 g — Table 1 - Microgel A activation: 0 g — Table 1 - Microgel A termination: 172 g — Table 1 - Microgel B butadiene: 13742 g — Table 1 - Microgel B styrene: 251 g — Table 1 - Microgel B TMPTMA: 125 g — Table 1 - Microgel B HEMA: 137 g — Table 1 - Microgel B water: 250 g — Table 1 - Microgel B initiation: 2740.5 g — Table 1 - Microgel B activation: 304.5 g — Table 1 - Microgel B termination: 105 g — Table 1 - Microgel A Tg: -75° C. — Table 2 - Microgel B Tg: -65° C. — Table 2 - Microgel A diameter d10: 43.5 nm — Table 2 - Microgel A diameter dz: 50.9 nm — Table 2 - Microgel A diameter d80: 55.7 nm — Table 2 - Microgel A specific surface area: 127 m2/g — Table 2 - Microgel A gel content: 91.7% by wt. — Table 2 - Microgel A swelling index: 13.1 — Table 2 - Microgel A OH number: 33 mg KOH/g polymer — Table 2 - Microgel A acid number: 7.4 mg KOH/g polymer — Table 2 - Microgel B diameter d10: 35.4 nm — Table 2 - Microgel B diameter dz: 48.2 nm — Table 2 - Microgel B diameter d80: 55.1 nm — Table 2 - Microgel B specific surface area: 139 m2/g — Table 2 - Microgel B gel content: 94.8% by wt. — Table 2 - Microgel B swelling index: 7.4 — Table 2 - Microgel B OH number: 37.8 mg KOH/g polymer — Table 2 - Microgel B acid number: 8.6 mg KOH/g polymer — Table 2 - HNBR acrylonitrile content: 34% by weight — Example 1 - HNBR residual double bond content: 3.5% — Example 1 - Mooney viscosity: 77 ML (1+4) @100° C. — Example 1 material description - Mooney viscosity: 40 ME — Table 4 - Mooney relaxation: 6.7% — Table 4 - Mooney viscosity: 37 ME — Table 4 - Mooney relaxation: 6.9% — Table 4 - Mooney viscosity: 32 ME — Table 4 - Mooney relaxation: 7.4% — Table 4 - Mooney viscosity: 39 ME — Table 4 - Mooney relaxation: 6% — Table 4 - Mooney viscosity: 42 ME — Table 4 - Mooney relaxation: 7.4% — Table 4 - Vulcanization temperature: 180° C. — Table 5 - Vulcanization test method: ASTM D5289 — Table 5 - Vulcanization pressure: 170 bar — vulcanization step - Vulcanization time: 9 min — vulcanization step - Compression set test compression: 25% — discussion - Compression set storage temperature: -30°C — discussion - Compression set storage temperature: 150°C — discussion - Compression set storage time: 70 h — discussion - Compression set storage time: 158 h — discussion - Compression set storage time: 168 h — discussion

Relevant Experimental Evidence - Preparation examples: microgels A and B were made by emulsion polymerization and characterized by Tg, size, gel content, and acid/OH numbers. - Examples 1-3: HNBR compositions without microgels were compared against inventive microgel-containing mixtures. - Examples 4-5: inventive mixtures showed comparable Mooney

behavior and vulcanization performance while improving final properties. - The patent demonstrates that microgels can improve high-temperature mechanical performance and demolding behavior in HNBR vulcanizates.

Technical Relevance This patent is relevant because it addresses formulation of hydrogenated NBR for high-temperature service, a major application area. It shows how microgels and peroxide/unsaturated-carboxylic-acid systems can improve the performance of HNBR-based vulcanizates without changing the base polymer chemistry.

Patent 5

- Patent Number: EP2519350B1
- Patent Title: Highly active metathesis catalysis selective for romp and rcm
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses ruthenium metathesis catalysts and their use in metathesis reactions, including a method for making modified nitrile butadiene rubber by depolymerization. The examples mainly describe synthesis of Ru complexes, while the patent text explicitly extends the catalyst utility to NBR depolymerization and hydrogenation-related uses. - Ligand synthesis scale: 5.0 mmol — Examples 1-19 - Ligand synthesis temperature: 85°C — Example 1 - K₂CO₃: 1.5 eq — Example 1 - 2,4,6-Trivinyl-cyclotriboroxane pyridine complex: 0.5 eq — Example 1 - Pd(PPh₃)₄: 2% — Example 1 - Ru complex synthesis scale: 1.0 mmol — Example 1 - Ru complex synthesis temperature: 20-25°C — Example 1 - Ru complex reaction time: 30-60 min — Example 1 - DCM: 15 mL — Example 1 - Ligand 3a: 250 mg — Example 1 - Ru precursor 1b: 860 mg, 1.0 mmol — Example 1 - CuCl: 270 mg, 2.5 mmol, 2.5 eq — Example 1 - Ligand 3b yield: 91% — Example 2 - Ru complex 4b yield: 29% — Example 2 - Ligand 3c yield: 54% — Example 3 - Ru complex 4c yield: 5% — Example 3 - Ligand 3d yield: 62% — Example 4 - Ru complex 4d yield: 32% — Example 4 - Ru complex 4e yield: 35% — Example 5 - Ru complex 4f yield: 14% — Example 6 - Ligand 3g yield: 56% — Example 7 - Ru complex 4g yield: 8% — Example 7 - Ligand 3h yield: 32% — Example 8 - Ru complex 4h yield: 42% — Example 8 - Ligand 3j yield: 93% — Example 9 - Ru complex 4j yield: 48% — Example 9 - Ligand 3k yield: 83% — Example 10 - Ru complex 4k yield: 68% — Example 10 - Ligand 3m yield: 49% — Example 11 - Ru complex 4m yield: 52% — Example 11 - Ligand 3n yield: 45% — Example 12 - Ru complex 4n yield: 64% — Example 12 - Ligand 3p yield: 67% — Example 13 - Ru complex 4p yield: 26% — Example 13 - Ligand 3q yield: 46% — Example 14 - Ru complex 4q yield: 38% — Example 14 - Ligand 3r yield: 44% — Example 15 - Ru complex 4r yield: 14% — Example 15 - Ligand 3s yield: 51% — Example 16 - Ru complex 4s yield: 85% — Example 16 - Ligand 3t yield: 38% — Example 17 - Ru complex 4t yield: 23% — Example 17 - Ligand 3u yield: 28% — Example 18 - Ru complex 4u yield: 32% — Example 18 -

Ligand 3v yield: 81% — Example 19 - Ru complex 4v yield: 10% — Example 19 - NBR depolymerization temperature: 30-100°C — claims/description - NBR depolymerization with mixed catalysts: 30-100°C — description

Relevant Experimental Evidence - Examples 1-19: a series of Ru complexes were synthesized and characterized, establishing the catalyst platform. - The patent explicitly claims use of the catalysts for making modified NBR by depolymerization. - The disclosure also states use in hydrogenation of rubbers containing carbon-carbon double bonds, linking the catalyst family to HNBR-related processing.

Technical Relevance This patent is relevant because it provides metathesis catalyst chemistry directly applicable to NBR modification before hydrogenation. It supports the broader HNBR synthesis toolbox by identifying catalyst structures and operating windows for depolymerization of nitrile rubber.

Patent 6

- Patent Number: US8957161B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses metathetic degradation of nitrile rubber in chlorobenzene using ruthenium catalysts and salt additives, followed by analysis of molecular-weight reduction. The process is carried out at room temperature with 1-hexene as co-olefin, and the patent shows that salts can accelerate or alter the efficiency of NBR depolymerization. - Acrylonitrile content: 35% by weight — General method - Mooney viscosity: 34 Mooney units — General method - Residual moisture content: 1.8% by weight — General method - Mw: 240 000 g/mol — General method - Mn: 100 000 g/mol — General method - PDI: 2.4 — General method - Solvent mass: 293.3 g chlorobenzene — General method - NBR mass: 40 g — General method - 1-hexene: 0.8 g (2 phr) — General method - Catalyst solution solvent: 20 g MCB — General method - Reaction temperature: room temperature — General method - Catalyst amount: 20 mg — Table 1 - Catalyst loading: 0.05 phr — Table 1 - Ru amount: 60 ppm — Table 1 - LiBr amount: 2 phr — Trial 1.02 - LiCl amount: 2 phr — Trial 1.03 - LiI amount: 2 phr — Trial 1.04 - CaCl₂ amount: 0.01 phr — Trial 1.05 - CaCl₂ amount: 0.05 phr — Trial 1.06 - CaCl₂ amount: 0.1 phr — Trial 1.07 - CaCl₂ amount: 0.5 phr — Trial 1.09 - CaCl₂ amount: 1 phr — Trial 1.09 - CaCl₂ amount: 2 phr — Trial 1.10 - CaCl₂ amount: 5 phr — Trial 1.11 - CaCl₂ amount: 10 phr — Trial 1.12 - Reaction time: 0, 30, 60, 90, 180, 420 min — Table 1/2 - Mw: 186 832 to 53 852 kg/mol — Trial 1.01 over time - Mn: 60 000 to 29 314 kg/mol — Trial 1.01 over time - PDI: 3.1 to 1.8 — Trial 1.01 over time - Mw: 186 832 to 51 827 kg/mol — Trial 1.02

over time - Mn: 60 000 to 28 793 kg/mol — Trial 1.02 over time - PDI: 3.1 to 1.8 — Trial 1.02 over time - Mw: 186 832 to 21 337 kg/mol — Trial 1.03 over time - Mn: 60 000 to 13 153 kg/mol — Trial 1.03 over time - PDI: 3.1 to 1.6 — Trial 1.03 over time - Mw: 186 832 to 51 827 kg/mol — Trial 1.04 over time - Mn: 60 000 to 28 793 kg/mol — Trial 1.04 over time - PDI: 3.1 to 1.8 — Trial 1.04 over time

Relevant Experimental Evidence - Trials 1.01-1.04: compared Grubbs II alone with LiBr, LiCl, and LiI, showing salt-dependent changes in degradation efficiency. - Trials 1.05-1.12: CaCl₂ was tested over a wide loading range, demonstrating the effect of alkaline-earth chloride on metathesis. - The data show that salt additives can strongly influence molecular-weight reduction of NBR, which is relevant to HNBR precursor engineering.

Technical Relevance This patent is relevant because it directly addresses metathesis degradation of NBR, a precursor step for low-viscosity HNBR. It provides evidence that salt additives can tune catalyst activity and molecular-weight reduction, which is important for controlling final HNBR processability.

Patent 7

- Patent Number: US7737233B2
- Patent Title: Catalyst systems and their use for metathesis reactions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses catalyst systems for metathesis reactions that combine a metathesis catalyst with an alkaline earth metal chloride, especially calcium chloride or magnesium chloride. It is explicitly applicable to metathesis of nitrile rubber, supporting controlled depolymerization of NBR prior to hydrogenation. - Salt:catalyst ratio: 0.01:1 to 10000:1 — claim 1 - Calcium chloride or magnesium chloride — claim 2 - Catalyst amount: 20 mg — Table 2 - Catalyst loading: 0.05 phr — Table 2 - Grubbs II catalyst MW: 848.33 g/mol — examples - Hoveyda catalyst MW: 626.14 g/mol — examples - Grela catalyst MW: 617.13 g/mol — examples - Acrylonitrile content: 34% by weight — general method - Mooney viscosity: 31 Mooney units — general method - Residual moisture content: 1.2% by weight — general method - Mw: 186.832 g/mol — general method - Mn: 60000 g/mol — general method - PDI: 3.1 — general method - Solvent: 293.3 g chlorobenzene — general method - NBR mass: 40 g — general method - 1-hexene: 0.8 g (2 phr) — general method - Reaction temperature: room temperature — general method - Catalyst solution solvent: 20 g MCB — general method - LiBr amount: 2 phr — comparative example - LiCl amount: 2 phr — comparative example - LiI amount: 2 phr — comparative example - CaCl₂ amount: 0.01 phr — example 1.05 - CaCl₂ amount: 0.05 phr — example 1.06 - CaCl₂ amount: 0.1 phr — example 1.07 - CaCl₂ amount: 0.5 phr —

example 1.09 - CaCl₂ amount: 1 phr — example 1.09 - CaCl₂ amount: 2 phr — example 1.10 - CaCl₂ amount: 5 phr — example 1.11 - CaCl₂ amount: 10 phr — example 1.12

Relevant Experimental Evidence - Comparative examples: LiBr, LiCl, and LiI were tested with Grubbs II catalyst. - Examples 1.05-1.12: CaCl₂ was evaluated across 0.01 to 10 phr, showing the claimed salt-acceleration effect. - The patent establishes alkaline-earth chloride as a useful additive for NBR metathesis, relevant to HNBR precursor reduction.

Technical Relevance This patent is relevant because it provides a catalyst-system strategy for accelerating NBR metathesis, a key precursor step in HNBR manufacture. The salt-additive concept can reduce catalyst demand or improve depolymerization efficiency before hydrogenation.

Patent 8

- Patent Number: US5569719A
- Patent Title: Vulcanizable partially hydrogenated nitrile rubber composition containing ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 1996
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses partial hydrogenation of acrylonitrile-butadiene copolymer rubber in methyl isobutyl ketone using a Pd/SiO₂ catalyst to obtain HNBR with controlled iodine value. The resulting partially hydrogenated rubber is then compounded and sulfur-vulcanized, with tellurium dithiocarbamate used as a vulcanization promoter in some formulations. - Bound acrylonitrile content: 36% by weight — Example 1 - Hydrogenation solvent: methyl isobutyl ketone — Example 1 - Catalyst: Pd/SiO₂ — Example 1 - Iodine value: 15 — Example 1 - Hydrogenation degree: 95% — Example 1 - Zinc oxide: 5 parts — Table 1 - Stearic acid: 1 part — Table 1 - FEF carbon black: 45 parts — Table 1 - Sulfur: 0.5, 0.75, 1.0 parts — Table 1 - Tetramethylthiuramdisulfide: 1.5 parts — Table 1 - 2-mercaptobenzothiazole: 0.5 parts — Table 1 - Tellurium diethyldithiocarbamate: 1.0 or 2.0 parts — Table 1 - Vulcanization temperature: 160° C. — Example 1 - Vulcanization time: 20 minutes — Example 1 - Sheet thickness: 2 mm — Example 1 - Iodine value: 10, 20, 30 — Example 2 - Tensile strength: 300, 284, 241 kgf/cm² — Example 2 - Elongation: 660, 630, 500% — Example 2 - Tensile stress at 100% elongation: 82, 89, 107 kgf/cm² — Example 2 - Hardness: 68, 68, 69 JIS A — Example 2 - Heat build-up: 40, 27, 19 °C — Example 2

Relevant Experimental Evidence - Example 1: HNBR with iodine value 15 and 95% hydrogenation was prepared and compounded into sulfur-vulcanizable recipes. - Example 1: tellurium dithiocarbamate-containing formulations showed improved tensile and heat-build-up behavior versus controls. - Example 2: HNBR iodine value was varied from 10 to 30, showing the effect of hydrogenation level on heat build-up and mechanical properties. - Example 2: lower

iodine value generally reduced heat build-up, supporting the importance of residual unsaturation control.

Technical Relevance This patent is directly relevant because it discloses the hydrogenation step used to make HNBR from NBR and links hydrogenation level to final vulcanizate performance. It also shows that partial hydrogenation around 95% can deliver useful balance of mechanical strength and heat resistance.

Patent 9

- Patent Number: US7846995B2
- Patent Title: ... -containing catalyst residues from optionally hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent focuses on removing ruthenium-containing catalyst residues from optionally hydrogenated nitrile rubber solutions using functionalized ion-exchange resins. The HNBR starting material is described as having been made by NBR metathesis with Grubbs II catalyst followed by hydrogenation with $\text{RhCl}(\text{PPh}_3)_3$, and the resin treatment is used to lower residual Ru content. - Acrylonitrile content: 34% b.w. — starting HNBR description - Residual double bonds: less than 0.9% — starting HNBR description - Mooney viscosity: about 40 ML 1+4@100° C. — starting HNBR description - Standard rubber solution concentration: 6.0% by weight — starting material - Resin mass: 0.5 g — Example 1A - Rubber solution mass: 180 g — Example 1A - Reaction temperature: ca. 100° C. — Example 1A - Reaction time: 66 hours — Example 1A - Drying temperature: 60° C. — Example 1A - Initial Ru content: 15 ppm — Table 1/2 - Ru content after treatment: 10 ppm — Table 1/2 - Ru removal: 33% — Table 1/2

Relevant Experimental Evidence - Example 1A: thiourea-functionalized resin reduced Ru content from 15 ppm to 10 ppm. - Example 2A: amine-functionalized resin gave the same 33% Ru removal as the thiourea resin. - The patent demonstrates a practical purification step for HNBR solutions after metathesis/hydrogenation processing.

Technical Relevance This patent is relevant because residual catalyst removal is an important finishing step in HNBR manufacture, especially for metathesis-derived materials. It provides a route to lower Ru content in hydrogenated nitrile rubber solutions, improving product purity and potentially downstream stability.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7919563B2 and US7772328B2 both disclose solution-phase NBR metathesis followed by hydrogenation, using monochlorobenzene, ethylene or 1-hexene co-olefins, and high-pressure hydrogenation with Wilkinson-type catalyst systems.
- Both low-molecular-weight HNBR patents report the same substrate composition basis: 34 mol % acrylonitrile NBR with Mooney viscosity 35, but they differ in emphasis on shaped-article utility versus polymer-property optimization.
- EP2580260A1 shifts from synthesis to formulation, using partially hydrogenated NBR in silicone blends and emphasizing low-solvent processing and compression-set improvement rather than precursor depolymerization.
- US9321910B2 and US5569719A both focus on compounding/vulcanization of HNBR, but US9321910B2 uses microgels and peroxide/unsaturated-carboxylic-acid systems, whereas US5569719A uses sulfur vulcanization with tellurium dithiocarbamate promoter.
- US8957161B2 and US7737233B2 both address metathesis degradation of NBR with ruthenium catalysts; US7737233B2 specifically claims alkaline-earth chloride additives, while US8957161B2 compares multiple salts including LiBr, LiCl, LiI, and CaCl₂.
- US7846995B2 is distinct in targeting post-reaction purification, showing that HNBR made by metathesis/hydrogenation can be treated with functionalized ion-exchange resins to reduce residual Ru.
- Across the metathesis-focused patents, solution chlorobenzene/monochlorobenzene is the dominant solvent, room temperature or 80°C are common metathesis conditions, and catalyst loading is typically expressed in phr or mg scale.
- Hydrogenation-specific disclosure is strongest in US5569719A and the low-molecular-weight HNBR patents, where hydrogenation level is linked to iodine value, residual double bonds, and final Mooney viscosity.

4. CONCLUSION

The patent set shows that hydrogenated NBR is most effectively accessed through controlled precursor engineering of NBR by metathesis, followed by hydrogenation and, where needed, purification or formulation tailoring. The strongest synthesis evidence supports solution metathesis of NBR in monochlorobenzene/chlorobenzene with ruthenium carbene catalysts, co-olefins such as ethylene or 1-hexene, and subsequent hydrogenation under high H₂ pressure using Wilkinson-type catalysts. For property control, the evidence favors low residual unsaturation, lower Mooney viscosity, and narrower molecular-weight distribution; in the formulation space, partial hydrogenation levels around 95% and careful promoter/additive selection improve heat build-up, compression set, and high-temperature durability. Recommended evidence-based synthesis parameters are: metathesis in monochlorobenzene at about 80° C. or room temperature with

controlled catalyst loading, hydrogenation at 1200 psi H₂ and 138° C. when using homogeneous Rh catalysts, and post-treatment to reduce residual Ru when catalyst carryover is a concern.

5. REFERENCES

- US7919563B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7919563B2/en>
- US7772328B2 | Low molecular weight hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7772328B2/en>
- EP2580260A1 | Blends of partially hydrogenated nitrile rubber and silicone rubber, ... | Not disclosed | EP | 2013 | <https://patents.google.com/patent/EP2580260A1/en>
- US9321910B2 | Microgel-containing vulcanizable composition based on hydrogenated nitrile ... | Not disclosed | US | 2016 | <https://patents.google.com/patent/US9321910B2/en>
- EP2519350B1 | Highly active metathesis catalysis selective for romp and rcm | Not disclosed | EP | 2019 | <https://patents.google.com/patent/EP2519350B1/en>
- US8957161B2 | Process for reacting chemical compounds in the presence of catalyst systems | Not disclosed | US | 2015 | <https://patents.google.com/patent/US8957161B2/en>
- US7737233B2 | Catalyst systems and their use for metathesis reactions | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7737233B2/en>
- US5569719A | Vulcanizable partially hydrogenated nitrile rubber composition containing ... | Not disclosed | US | 1996 | <https://patents.google.com/patent/US5569719A/en>
- US7846995B2 | ... -containing catalyst residues from optionally hydrogenated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7846995B2/en>